Research on risk quantification and management in enterprise R&D activities

Xiaoyan Guo
Party School of the CPC Huangshi Municipal Committee
Huangshi, Hubei, China
406656543@qq.com

Abstract

Technological innovation is one of the most important means for enterprises to create core competitiveness, and R&D is an important link of technical innovation. Therefore the risks of R&D have a direct relationship with the enterprises' R&D activities proceeding smoothly. In this paper, using the mean-variance model of classic portfolio theory to research on risks of R&D quantitatively and qualitatively, and building the enterprise risk value, risk tolerance and risk degree of three risk indexes. It is helpful for enterprises to make corresponding risk decision and get the Optimal Investment Portfolio of R&D assets. Finally, put forward the management countermeasures of enterprise R&D risks from the enterprise itself and the macroscopic angle.

CCS Concepts

• **General and reference** → Cross-computing tools and techniques; Design; • **Computing methodologies** → Modeling and simulation; Model development and analysis; Modeling methodologies; • **Applied computing** → Enterprise computing; Reference models.

Keywords

Technology innovation, Mean-Variance model, R&D risk, Risk management

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1 INTRODUCTION

With the rise of knowledge-based society, science and technology has become the decisive force of economic and social development. As the main body of market activities, enterprises must become the basic force to promote the construction of innovative society. R&D (research and development) is the main way of technological progress, the premise of survival and development, and an effective means to improve labor productivity, competitiveness and economic benefits. The key to the success of R&D and R&D is the formation of high R&D investment and its sustainable development, However, high investment often brings high risk, that is, there is great uncertainty in the future income, which will bring certain risks to enterprises.

As R&D activities are the most characteristic and dynamic part of enterprises, and accompanied by high risk, the measurement and management of R&D risk has became the focus of domestic and foreign experts and scholars. In 1961, Cramer proposed a quarter-interval method to measure risk because the distance of individual single-valued discrete expected mean had too great influence on risk measurement [1]. Konno and Yamazaki also put forward the idea of absolute deviation to measure risk [2]. Yu Liaoyuan et al. put forward a fuzzy comprehensive evaluation method of failure mode based on extension to quantitatively evaluate the project risk of chip development [3]. Li R. M et al. took into account the Modular rocket of product network and organizational network in CoPS (Complex Products and Systems) project, met Al constructed a product-organization CoPS interdependent network, and established a dynamic model of technological risk diffusion in complex product R & D process, this paper simulates the technology risk diffusion caused by a few tasks, and then discusses the influence of risk diffusion caused by a few component failures and design changes in interdependent networks, three risk mitigation strategies and their combination effects were estimated by numerical simulation, and the network robustness was evaluated by attacking the above-mentioned networks on the basis of the gray attack scenario [4]. Han Guanghui proposed that KMV model had some validity in credit risk measurement and could be used as risk early warning [5]. Yan Haibo's research found that incorporating ESG (Environment, Social Responsibility, Corporate Governance) factors into portfolio construction can reasonably avoid risks, that was, ESG portfolio had the function of stable returns and risk avoidance, and the higher the ESG rating, the lower the risk and the higher the return [6]. Zhang Junxia et al. adopted the valuation weighting method to establish the risk assessment method of science and technology research and development before and during lending, and established the risk assessment framework of financial support for science and technology innovation, providing a measurement method for the specific implementation of financial institutions' support for science and technology innovation [7].

Tummala believed that R&D project risk management system consists of risk identification, risk measurement, risk evaluation and risk monitoring [8]. In 2000, the Project Management Association (PMD) revised the project risk management, the new project risk management included risk management plan, risk identification,

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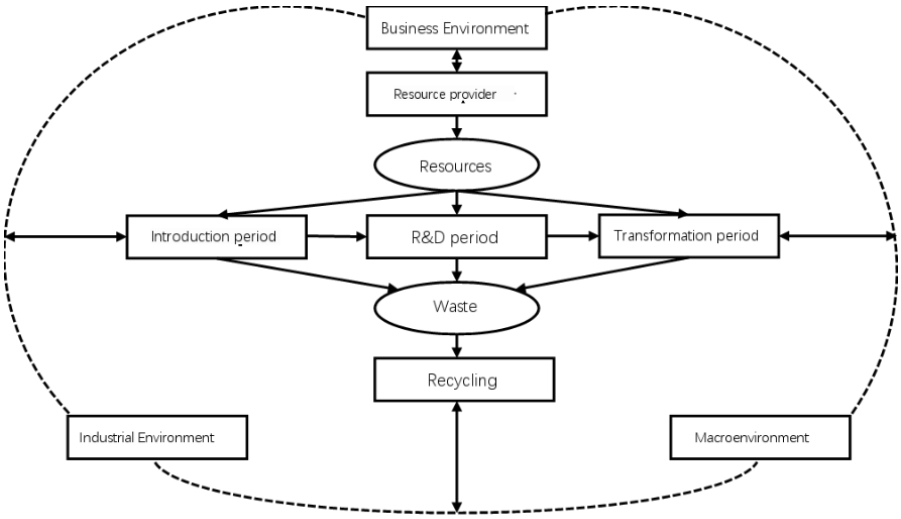

Figure 1: Enterprise R&D System Diagram.

risk qualitative analysis, risk Quantitative analysis, risk response plan, risk management and control, adding qualitative analysis and Quantitative analysis to enterprise risk management, and finally controlling risk management, made the risk management process more concrete. Gu Xiaoyan et al. proposed that there were many risk factors and complex relationships affecting the development of complex product systems, which were mainly concentrated in the three stages of research, design and manufacturing [9]. Tan Ling pointed out that research and development risks mainly include two points: first, the shortage of scientific and technological personnel and the backward level of technical equipment would have an impact on the transformation of scientific and technological achievements of scientific and technological enterprises, which would lead to scientific and technological products could not give full play to their due value, and then cause relative waste; Second, the continuous growth of research and development expenditure would lead to increased financial risks in the short term [10].

Through a large number of literature review, it was found that the research on R&D risk at home and abroad mainly focuses on the innovation of risk management methods and the evaluation of R&D risk. Most of the research methods are qualitative analysis, and it was common to establish risk evaluation model, but there was little research on the measurement of R&D risk. Based on this, the study focuses on the measurement model of enterprise R&D risk, and puts forward countermeasures for R&D risk management.

2 FORMATION MECHANISM OF ENTERPRISE R&D RISKS

A survey in the United States showed that only about 60% of the R & D plans of high-tech Enterprises were technically successful, while only 30% of the R & D plans that were technically successful could be marketed, only 12 percent of high-tech products brought to market were ultimately successful. R&D risks refer to the possibility of loss caused by the difference between the actual results and the expected results of R&D activities due to the uncertainty of various environmental factors, the complexity of R&D projects and the limited innovation ability of R&D personnel.

From the perspective of the process of R&D activities, Brown and Sevenson used the process system theory to divide the R&D process into five stages: input, process system, output, enterprise acceptance system and implementation results [11]. Each link of R&D activities is not a simple sequential connection. The activity itself is in the exogenous environment and keeps close interaction with the exogenous environment, the improper disposal of these exogenous environmental factors will have an impact on R&D activities. According to the level of exogenous environment, Yu Liang and Lu Zhenghua divided the exogenous environmental factors affecting R&D activities into three categories: enterprise environmental factors, industrial environmental factors and macro environmental factors [12]. On the basis of these studies, this paper divides the activities into three stages: introduction period, R&D period and achievement transformation period, and constructs the R&D system diagram of R&D activities, as shown in Figure 1.

As the most important activity for enterprises to cultivate core technologies and products, R&D is bound to be in the unique environment of enterprises. The smooth development of R&D activities is affected by the investment of funds, the quality of personnel, the advanced level of equipment, the support of enterprise organizations for R&D, team cooperation and innovation ability, information acquisition and communication, etc.

The industrial environment factors faced by enterprises include industrial scale, industrial layout, competition status, entry barriers, etc., which greatly affect the effect of R&D.

All the macro-economic activities, such as R&D, the macro-economic situation of enterprises are affected by R&D policies and macro-economic environment, including the macro-economic situation of enterprises.

It can also be seen from Figure 1 that the R&D risk of an enterprise not only comes from the exogenous environmental risk of R&D activities, but also the internal relationships, including the

timely and sufficient supply of resources, and the disposal of waste products. If these relationships are not properly handled, it will have a very adverse impact on R&D.

3 RISK MEASUREMENT OF ENTERPRISE R&D ACTIVITIES

From the above analysis of the formation mechanism of enterprise R&D risks, we can see that the R&D activities of enterprises face different internal and external risks in nature, source and degree, so they have to diversify risk portfolio including R&D equipment, capital, personnel and system construction. Facing the reality of asset management, we can use the mean variance model of portfolio theory proposed by American economist and Nobel laureate Harry M. Markowitz in 1952 to measure the R&D risks of enterprises. Although the mean variance model is mainly used in the field of securities investment, R&D activities can also be regarded as a kind of long-term investment, which can obtain corresponding returns through capital investment, Therefore, R&D risks should be measured by this model. The biggest characteristic of using mean variance model to measure R&D risks are that the influence and consequence of various risks in the R&D process are borne by the final R&D investment return rate, which does not need to consider the impact of internal and external risks of the enterprise separately, which reduces the difficulty of measurement.

3.1 Value at risk of R&D

The R&D investment is divided into equipment investment (I_1), capital investment (I_2) and human resources investment (I_3). Assuming that the total R&D investment is $\sum I_i$, and the proportion of each investment in the total investment is W_i, then $\sum W_i = 1$. If the possible return on investment (ROI) of each investment is R_i (in the above formulas $i = 1,2,3,\ldots n$), then the expected return on investment of R&D investment of the enterprise $\hat{R} = \sum R_i * W_i$ ◦ At this time, the risk of R&D investment is the deviation degree between the possible return and the expected return, and the variance of the expected return can be used σ^2 to measure:

$$\sigma^2 = \sum \left(R_i - \hat{R}\right)^2 * W_i \tag{1}$$

It can be seen from equation 1) that if the expected return on investment of R&D activities $\hat{R}$ under certain circumstances, the optimal portfolio is the investment portfolio which can minimize the risk of enterprise R&D investment. However, risk management is not simply to minimize the risk, but to maximize the return on the basis of taking a certain amount of risk, therefore, the risk tolerance and risk degree of subsequent research and development should be analyzed and studied.

3.2 Measurement of R&D risk tolerance of enterprises

In the process of R&D risk management, enterprises often face the situation that before R&D activities are carried out, enterprises will put forward several R&D schemes, and there is no strict distinction between right and wrong. If the enterprise is not determined at the beginning whether it is willing to take or evade the risk, and how much risk it is willing to take, no matter which R&D plan is adopted, it will make enterprise managers or R&D subjects in a dilemma. Therefore, the enterprise R&D activities should be clear before the risk tolerance.

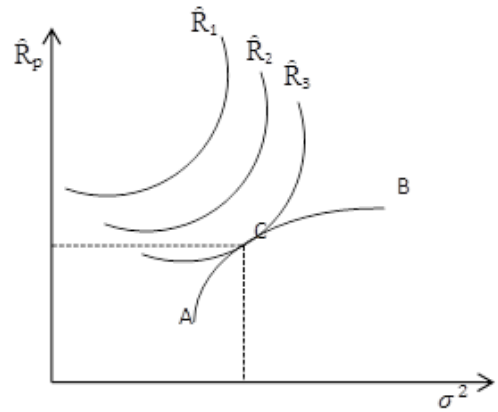

Figure 2: Mean variance indifference curve of investors.

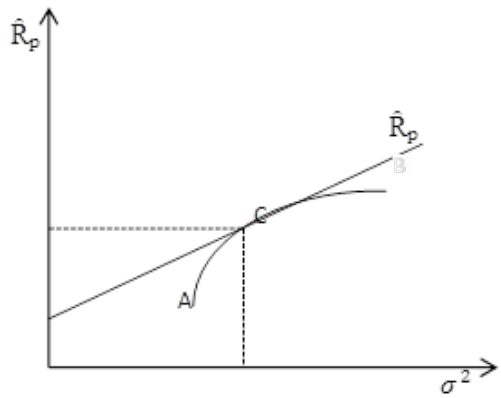

Figure 3: Approximate indifference curve model.

This paper still uses the mean variance model of classical portfolio theory of Harry m Markowitz to measure R&D risk tolerance. According to the risks of R&D asset portfolio, the assets can be divided into risk asset portfolio and risk-free asset portfolio. Assuming that there are risk-free assets, the effective set of asset portfolio and indifference curve of R&D investors are given, then the optimal portfolio selection is to find the portfolio with the greatest utility in the intersection set of R&D investment portfolio. As shown in Figure 2.

On the mean-variance plane, the slope of the investor indifference curve is positive, the indifference curve is shown as $\hat{R}_1$, $\hat{R}_2$, $\hat{R}_3$ in Figure 2. The slope of the indifference curve represents the substitution rate between risk and return. The higher the slope is, the higher the income compensation that must be provided to investors in order to let them take more risks, which indicates that the investors are more risk averse. Again, the smaller the slope,

the lower the degree of risk aversion. The curve AB in Figure 2 is the efficient set of the portfolio, and the efficient set is a right upward sloping curve, which reflects the characteristics of "high risk, high return". Point C is the frontier asset portfolio with the optimal risk asset portfolio, and the expected return on demand is strictly higher than that of the minimum variance portfolio.

According to the above theory, we can estimate the R&D risk tolerance: once a certain portfolio is selected, the indifference curve should be tangent to the effective set of the portfolio, so the slope of the two is the same. So, in a tiny range, you can use a straight line $\hat{R} = a + \frac{1}{b}\sigma^2$ to approximately replace the indifference curve, as shown in Figure 3.

Straight line $\hat{R}$=a+$\frac{1}{b}\sigma^2$Medium, $\hat{R}$ is the expected return rate of the risk-free portfolio and the risk-free portfolio, a is the intercept of an approximate indifference curve on the longitudinal axis, $\frac{1}{b}$ is the slope of the indifference curve, that is, the substitution rate between risk and return, then b is the risk tolerance.

Let X_S be the proportion of risk asset portfolio in enterprise R&D portfolio, and $(1 - X_S)$ be the proportion of risk-free asset portfolio. Then the expected return of the total asset portfolio is:

$$\hat{R}_p = X_S R_S + (1 - X_S) R_f \tag{2}$$

Among them, R_s is the expected rate of return of the company's risk assets portfolio, and R_f is the return rate of risk-free assets

From formula (2), it can be concluded that:

$$X_S = \frac{\hat{R}_p - R_f}{R_s - R_f} \tag{3}$$

Variance of portfolio:

$$\sigma_p^2 = X_s^2 \sigma_s^2 \ + (1 - X_S)^2 \sigma_f^2 + 2X_S (1 - X_S) \sigma_{sf} \tag{4}$$

Among them, $\sigma_s^2, \sigma_f^2, \sigma_{sf}$ are the variance of risk portfolio, the variance of risk-free portfolio, and the covariance of risky portfolio and non-risk portfolio. Since F is a risk-free portfolio, then $\sigma_f^2 = 0, \sigma_{sf} = 0$Therefore, equation 4) can be simplified as σ_p^2=$X_s^2 \sigma_s^2$, by substituting formula (3), we can get the following results:

$$\sigma_p^2 = \frac{\left(\hat{R}_p - R_f\right)^2 \sigma_s^2}{\left(R_s - R_f\right)^2} \tag{5}$$

Equation 5) can be regarded as the relationship between expected return and portfolio variance between risk asset portfolio and risk-free asset portfolio, which constitutes the feasible set of all risk assets and risk-free asset portfolio. According to figure 3, the slope of the point on the curve represented by combination P is:

$$\frac{d\sigma_p^2}{d\hat{R}p} = \frac{2\left(\hat{R}_p - R_f\right)\sigma_s^2}{\left(R_s - R_f\right)^2} \tag{6}$$

So that, $\frac{d\hat{R}p}{d\sigma_p^2} = \frac{(R_s - R_f)^2}{2(\hat{R}_p - R_f)\sigma_s^2}$ is the tangent slope of point C of the optimal portfolio. Since the indifference curve is tangent to point C, the slope of the indifference curve is equal to the tangent slope of point C. Namely: $\frac{1}{b} = \frac{(R_s - R_f)^2}{2(\hat{R}_p - R_f)\sigma_s^2}$=

$$b = \frac{2\left(\hat{R}_p - R_f\right)\sigma_s^2}{\left(R_s - R_f\right)^2} \tag{7}$$

By introducing formula (3) into (7), the risk tolerance of enterprise R&D is obtained:

$$b = \frac{2X_S \sigma_s^2}{R_s - R_f} \tag{8}$$

It can be seen from formula (8) that the greater the proportion of R&D activities invested in risk assets, the higher the risk tolerance required by enterprise R&D.

3.3 Construction of R&D risk degree

Xu Jiuping used the entropy value in information theory as a quantitative index to measure the uncertainty degree (investment risk) of investment scheme, and established the investment risk degree on the fixed value of risk compensation rate [13]. On the basis of this research, combined with the corresponding measurement model of R&D risk value and risk tolerance, this paper defines the degree of risk as follows: the ratio of R&D risk compensation rate to risk tolerance, i.e

$$DR = \frac{-\text{ risk tolerance}}{\text{Risk tolerance}}$$

It can be obtained from formula (1) and formula (8)

$$DR = \frac{\sigma^2 - b}{b} = \frac{\sum\left(R_i - \hat{R}\right)^2 * Wi - \frac{2X_S\sigma_s^2}{R_s - R_f}}{\frac{2X_S\sigma_s^2}{R_s - R_f}} \tag{9}$$

Then, when we know the R&D risk value of an enterprise and its risk tolerance, then we can calculate the R&D risk degree of the enterprise. Through investigation and statistics, we can infer a reasonable risk degree *DR* value, and set this reasonable risk degree value as *DRr*. If the DR value of an enterprise's R&D activity is far less than the *DRr* value, it can be put into the R&D activities; If the *DR* value of an enterprise is far greater than the value of *DRr*, then it is suggested that the enterprise should not carry out the research and development activities to avoid losses; If the *DR* value of an enterprise is close to *DRr*, then the enterprise should carefully consider whether to invest a large amount of assets in the R&D activities.

4 CONCLUSION

Risk management is a whole process including risk identification, risk estimation, risk evaluation, risk control and management effect evaluation. This is a more accurate and comprehensive definition in academic circles. As an important activity in enterprises, the risk control and management of R&D is the key to improve the competitiveness of enterprises.

4.1 Establish risk awareness and pay attention to risk prediction

Enterprise leaders, decision makers and relevant managers, especially the R&D subject, should fully understand all kinds of potential risk factors and possible losses. We should have a clear understanding of the risks in the process of R&D, reduce the subjective knowledge of risks, and use objective and rational scientific methods to predict and evaluate risks, so as to avoid or transfer risks.

4.2 Ensure adequate supply of R&D funds

R&D is a "high investment, high risk" activity. Enterprises need to invest funds strategically to ensure the capital needs in the R&D process. On the premise of not affecting the normal operation of other activities of the company, if there is surplus funds, special R&D funds can be established. If the company is short of funds, the introduction of venture capital industry is an effective way to reduce the risk of R&D, because venture capital itself is willing to bear the high risk of the investment object in order to obtain high returns.

4.3 Strengthening the informatization construction of enterprise R&D

R&D activities not only need to have excellent technical ability, but also need information along with the whole R&D process. As far as the enterprise is concerned, if we want to control the R&D departments, we need the information communication, sharing and feedback between the departments. Outside the enterprise, it is necessary to build an information system, quickly collect effective information resources, and maintain smooth contact with the market, accurate discovery and grasp of opportunities, thus greatly reducing the risk of R&D.

4.4 Making full use of the preferential policies given by the government

The government has given a lot of preferential policies to the overall development planning and regulation, technological innovation, talent attraction and training in the macro aspects of Taxation, finance, financing and policy. For example, the Innovation Fund for small and medium-sized science and technology enterprises, 15% income tax exemption for high-tech enterprises, subsidies for engineering technology centers, and subsidies for introducing talents are designed to improve the R&D capability of enterprises and reduce the risk of R&D. Therefore, enterprises should make full use of the preferential policies given by the government to provide assistance for R&D activities.

R&D activities are the core competitiveness of an enterprise. The smooth implementation of R&D process can improve the independent innovation ability of enterprises. Therefore, improving the level of enterprise research and development and the success rate of the project has become a key research topic of enterprises. This paper analyzes the formation mechanism and characteristics of R&D risk, and provides the basis for risk avoidance. In the process of R&D, the most important measure of R&D risk is to reduce the risk of R&D investment. Last, from the enterprise level and the government macro level, this paper puts forward the corresponding countermeasures for enterprise R&D risk management, so that enterprises can formulate corresponding risk prevention measures.

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